

Network computing

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Network computing represents a new computing paradigm in which all information, data and software applications reside on a network and are accessed on demand by users. This approach to computing offers the potential for users to access 'everything' from 'anywhere', removing the constraints of localised storage of information and applications, and, based on evolving standards, allowing access from a range of 'information appliances'. This paper introduces the basic ideas behind network computing, and examines a range of forms it could take. It describes the current status of commercial network computing approaches, with particular emphasis on the use of the Java language, and examines the associated benefits and technical issues. A prototype BT system, developed using Java client software and CORBA-based middleware, is described.

1. Introduction

Network computing represents a new computing paradigm in which all information, data and software applications reside on a network and are accessed on demand by users. This approach to computing offers the potential for users to access 'everything' from 'anywhere', removing the constraints of localised storage of information and applications, and of proprietary hardware and software architectures, provided the necessary standards can be established.

In the ultimate version of network computing, users accessing the network from a device with a minimum set of capabilities would potentially have access to unlimited resources of information and applications, provided they had a connection to the network of adequate performance, in terms of bandwidth, latency and reliability. The applications they could run would not be constrained by the characteristics of their access device; servers within the network would flexibly partition tasks between the user's access device and machines elsewhere on the network.

The currently prevailing view is that the main advantage of network computing is in reducing total cost of ownership for 'desktop' computers, particularly within corporate environments. While this reduced cost is likely to be true, it is largely as a consequence of costs created by specific hardware and software inadequacies of the currently dominant desktop machines, caused more by the commercial market in which they have evolved than underlying technological issues. However, in the longer term, the real advantages of network computing will be in the radical changes it brings in the flexibility and productivity of computing, leading, in the business

environment, to overall improvements in return on investments, and, in the wider context, to the mass-market use of 'computing' for a wide range of everyday activities.

The increasing use of the Internet, particularly the World Wide Web (WWW), represents the first steps on the most likely evolution path towards network computing, and it has begun to show the possibilities offered by ubiquitous access to information and services. For the potential of network computing to be fully realised many technological issues will need to be addressed, with the fundamental issue being the development and acceptance of the standards that will underpin network computing, and these are likely to evolve within the Internet standards arena.

2. The network computing model

Computing technology is now being seen to have occurred in three waves. The first wave, through the 1960s and 1970s, was host-centric, with dumb terminals connected via dedicated links to mainframe host computers. These host machines had time-shared operating systems and mainly ran large monolithic applications and databases, predominantly for business purposes. The second wave, through the 1980s and into the 1990s, was desktop-centric, with personal computers (PCs) — mainly IBM-compatible machines with Microsoft operating systems, and Apple Macs — being used to run a wide range of applications, with word processors and spreadsheets as the main 'productivity' software. These PCs were initially not networked, with data and applications being transferred and loaded on diskettes, but increasingly they were connected on to workgroup local area networks (LANs) to provide centralised file and print servers.

In the third wave, beginning in the early 1990s and now known as network-centric or network computing, the user's machine is connected to a network, or more generally an internetwork (an interconnected collection of networks), on which reside a range of server machines, which provide access to information, applications and services. In principle, all users on the network can access all the servers and all the other users' machines, enabling the communication of information between users. In addition, servers can access other servers, enabling one server to draw on applications or services they provide. Early client/server systems provided a limited form of network computing, but it was the WWW running on the Internet, the original internetwork, that saw the real beginning of network computing.

The concept of network computing involves an end-user interacting through input devices, e.g. keyboard and pointing device (mouse), and output devices, e.g. display screen and audio, with computing services and applications via a communications link. Figure 1 shows the possible stages involved in a typical computing activity, and the type of information passing across the interfaces between the stages. In principle, if any one or more of these interfaces is running over a communications link then network computing is taking place. Depending on where the links are introduced different network-computing models are produced, with different bandwidth and performance requirements on the links.

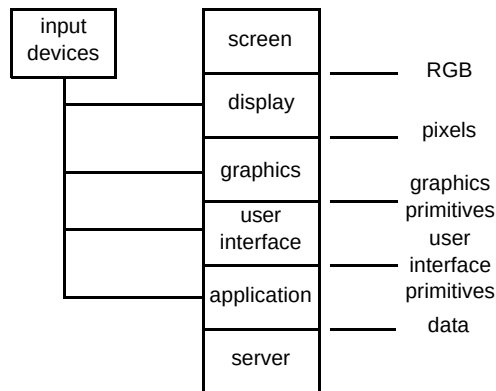


Fig 1 Stages in computing activity.

Putting the link at the analog video RGB interface, which corresponds to conventional broadcast or closed-circuit television, would require very high analog bandwidth and is not a practical option. A link at the pixel interface, which corresponds to digital television transmission, would require high digital bandwidth with low latency, but could use low complexity and low-cost end-user hardware. This produces a model that is closely related to interactive multimedia services ('video on demand') which are currently being proposed and trialled in

various places, and such services offer possibilities for adding more advanced network computing features.

Introducing a link at the graphics primitive interface corresponds to the conventional 'dumb' terminal model, which has been used on mainframe computers for many years. In this case the graphics primitives are mainly characters, which are then rendered into pixels on the terminal. In recent years graphical terminals have been introduced in which the graphics primitives are lines, shapes, windows, bit-maps, etc, in addition to characters. The most common form of graphical terminal, the X-terminal, which uses the X-Window protocol, has been widely used in some application areas, including finance trading floors. X-terminals, and others using a related proprietary protocol, ICA, are also now being used to access Microsoft Windows applications running on network servers.

In the above models the end user's machine is essentially a user interaction device, providing display and input facilities for an application running remotely. In the remaining models at least part of the application is running on the end user's machine. This part can be the user interface (presentation) logic, which collects (and possibly checks) user input and, following processing by the remote application, presents output results. The interface with the remainder of the application is at the level of the user interface primitives, such as dialogue boxes, menus, buttons, text boxes, etc. The forms input capabilities of the WWW represent a basic form of this model, with the use of browser-based scripting languages, such as JavaScript, providing more advanced features such as data entry checking.

The final model, which is what is commonly referred to by the term 'network computing', has a complete application, for example, a spreadsheet, running on the end user's machine, but the application is downloaded when required from a server. In addition, the application can access information and services on other servers on the network.

The above models could all provide an identical network computing capability to an end user, who, given equivalent display and input devices in different locations, for example office and home, could have access to identical information and services, and would not need to know where and how processing was taking place. The different computing models produce different trade-offs between complexity of end-user machines, performance of communication links, and the distribution of information and application servers. The overall costs of the different models are dependent on a range of factors, particularly the cost trends of different technologies, for example

processing power, volatile (RAM) and persistent (disc) storage, and bandwidth, which themselves depend both on technological advances and future market conditions. However, before the type of ubiquitous network computing service referred to above becomes feasible, irrespective of the network computing model used, a number of significant advances in software and protocol technology need to be made first.

3. The benefits of network computing

The network computing model has a number of advantages when compared to the host-centric and desktop-centric approaches described above. For the end user the most substantial and perceptible benefit of network computing will be in ease of use, with two main aspects.

Network computing fundamentally shifts responsibility for issues such as installation, update, administration, reliability and availability from the end-user to the application and service providers on the network, in contrast to an infrastructure based on PCs, where individual end-users are primarily responsible for provision and support of as their own desktop. Provision of appropriate resources to meet the requirements of the end user is the responsibility of the application and service providers on the network, and as their resources are centralised they will be better able to exploit economies of scale and to provide cost-effectively the facilities (such as expertise and specialised equipment) required to provide a high-quality computing environment. Accordingly the network computing model has the potential to provide the high levels of availability and reliability which characterised the host-centric computing infrastructures, together with the usability and flexibility of the desktop-centric approach.

Also, ubiquitous access to a competitive and potentially global network of application and service providers made possible by the network computing model should ensure that the high degree of autonomy and freedom which PCs gave to the end user (and which made them so popular) continues to be enjoyed and that there is no return to the inflexibilities and restrictions which characterised the host-centric approach. The freedom PC users have to install any software required at the local level (thereby avoiding the rigid and unresponsive central IT systems of the past) is replaced in the network computing model by the ability to access the functionality required from a global network of supplier sites. This realignment of responsibilities engendered by network computing will be experienced by end users as 'ease of use' and by those paying for computing infrastructures as 'lower cost of ownership'.

The network computing model also has the potential to improve ease of use at the application level. The restricted

nature of the application delivery mechanism in the desktop-centric approach, via 'shrink-wrapped' software, has resulted in a very substantial increase in the size and complexity of PC applications. The limitations of software distribution in this form place obvious restrictions on the frequency of software updates and limit the extent to which applications can be customised to the requirements of each individual user. Thus each application has to contain every possible function any user might need, which has resulted in 'bloated' applications, requiring increasing memory sizes and faster processors. For users, particular functions can be difficult to find and use, and any given individual routinely uses only a small subset of the facilities available.

In contrast, the distribution mechanism for network computing applications, i.e. downloadable code and remote access to distributed applications, supports delivery of functionality to the user on demand. Application architectures based on fine-grained components are natural to this mode of delivery and, where bandwidth to the user is limited, it may be a requirement. Accordingly, it will be possible to organise application functionality so that users routinely only access and run the parts they require. Functions required only occasionally will be downloaded as necessary rather than bulking out the application.

4. Current developments

As mentioned above, in practical terms the current developments of network computing are taking place by extending the capabilities of the WWW. Initially the WWW provided access to static pages of information, with users moving between servers on the Internet by 'clicking' on hypertext links. However, the inclusion in the WWW by its inventor, Tim Berners-Lee, of a common gateway interface (CGI) protocol and a forms-input mechanism enabled it to rapidly evolve into a type of general computing platform, leading to the development of a wide range of novel Internet-based applications. The recognition that the browsers used to access WWW could be run on devices of lower complexity than current multimedia PCs led to the development of a range of prototype access devices, which represent the first examples of 'network computers'.

A specific profile of required capabilities for such an access device, known as the Network Computer TM or NCTM, has been proposed and supported by a powerful group of IT companies, including IBM, Sun, Oracle, Apple and Netscape. Different network computer specifications are emerging and being promoted by other companies, but they are broadly equivalent.

The NC Reference Profile [1] specifies a minimum hardware configuration (640 × 480 resolution screen, pointing device, text input capability, audio output), and

support for Internet network protocols (IP, TCP, UDP, SMTP, HTTP, etc), and WWW formats (HTML, GIF, WAV, etc). In addition it specifies that the NC must support the Java software environment, thus enabling Java software to be downloaded from servers to the NC as and when required, and discarded when no longer needed (although in practice it may be cached locally). The software downloaded will always be the latest version, it will configure itself on the NC automatically, and it can potentially be provided on a leased or pay-per-use basis.

Incorporation of a phone handset and keypad into an NC-based device could provide unified access to all communication and information facilities, including all advanced phone services, unified access and management of voice, e-mail and fax messages, and access to both personal and public directories.

In terms of underlying platform architectures all the major IT companies (including Microsoft) are either actively supporting or migrating towards the network computing model. For all companies other than Microsoft this architecture is based on Java [2] and CORBA (Common Object Request Broker Architecture [3]), but in all cases also provides support for ActiveX [4] and DCOM (Distributed Component Object Model [5]). For example, Netscape have announced that from 1997 their client environment (Netscape Communicator), server suite (Netscape SuiteSpot) and development environment (Netscape One) will support CORBA IIOP (Internet Inter-ORB Protocol) [6, 7]. Netscape's support for CORBA has encouraged a number of software toolmakers to develop products compliant with CORBA [8].

Similarly, Oracle have developed an infrastructure based on CORBA known as the Network Computing Architecture. The architecture is based on an Object Request Broker (ORB), called ICX, which provides interoperability between software 'cartridges', client applications and the extended-relational Oracle version 8 to be released in 1997 [9, 10]. Sun and IBM are also offering enterprise platforms based on CORBA ORBs to support their network-computing client hardware. Their ORBs are known as NEO and SOM respectively and, like Oracle, their strategy involves a migration path for their existing customer base in mainframe and client/server infrastructures to distributed object platforms.

Another potentially important participant in the development of current standards for network computing is the WWW Consortium (W3C) which has begun to consider the utility of integrating Object Management Group (OMG) and WWW standards, and OMG and W3C have become subscribing members of the each other's organisations [11, 12]. Microsoft has recently adopted a strongly Internet-focused approach to application delivery with the

development of ActiveX and DCOM. While this currently stops short of pure network computing there are indications that in order to address the issue of cost of ownership Microsoft may move towards a more explicit network computing approach based on the NetPC [13–15].

5. Java and network computing

The use of the programming language Java [2], and its associated Java virtual machine, opens up the possibility of applications that can run anywhere on the network, on client machines or servers, leading to the slogan 'write once, run anywhere'. In addition, the object-oriented nature of Java facilitates the concept of 'component-based' software, in which the range of aspects or features of applications are implemented as separate components. For example, most usage of word processing is to produce short, simple documents, and the software required can be simple and lightweight, and in this case would be downloaded to the NC. More advanced features would be provided by additional software components, which could be downloaded to the NC or, if appropriate, could be run on a server.

The key features of Java as a programming language for network computing are the Java application programming interfaces (APIs) and the Java virtual machine (VM). Every Java program is compiled to a machine-independent byte code that is, in effect, the instruction sequence for a virtual machine architecture abstracted away from any vendor specific processor. The Java byte code is then interpreted by the VM to convert the byte code into processor-specific instructions. Provided with every VM is a standard set of APIs providing a guaranteed set of capabilities, such as a windowing system, file I/O, networking, etc. Figure 2 is a typical architecture for an existing PC environment.

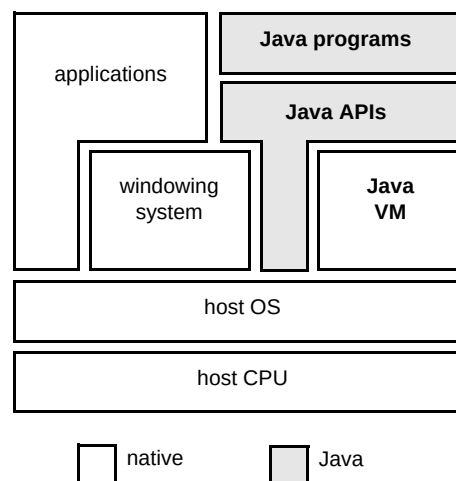


Fig 2 The Java environment architecture.

Java helps network computing through several of its key properties:

- platform independence — once each vendor has provided an implementation of the Java VM and APIs for their host architecture, the Java byte code will run without modification on any platform,
- code mobility — the ability to transmit small applications, applets, over the network, along with the HTML pages of the Internet, allows code to be run where it is needed and then discarded after use (this helps out with some of the versioning problems associated with software upgrades, as the software only needs to change once on the server),
- dynamic — Java is very new and while this brings some problems in terms of maturity it brings a host of benefits, e.g. due to the speed of adoption and the association with the Internet, the progression of Java has been very fast, with new APIs and tools being announced on a regular basis, and the power and flexibility of Java capabilities increasing rapidly,
- network aware — by its very design Java is network aware, having a range of networking technologies built into the core system and a large number of additional support facilities provided by third parties which provide its extensive interoperation capabilities,
- full and rich APIs — a key feature of any language is its support classes or APIs; Java provides all the basic feature sets required for a wide range of applications and has the ability to use more extensive systems on a dynamic basis.

6. Technical issues

The primary obstacles to the development of network computing are essentially commercial rather than technical. Key issues include the willingness and ability of independent software vendors (ISVs) to produce a wide enough range of network-computing applications to rival those available for PCs, and the speed with which the major proponents of network computing can deliver versions of their various products which support the new approach.

However, there are a number of technical issues which must be addressed before the full potential of the network computing model can be realised. Failure to resolve these issues promptly is likely to impede development of the network computing model over the long term. However, as network density, bandwidth, and reliability increase and costs fall, the advantages of using the network as a mechanism for delivery are likely to prove irresistible. The rate and manner in which these issues are resolved will strongly influence which of the various approaches to network computing currently being developed ultimately dominates the computing infrastructures of the future. Slow resolution of the issues would favour Microsoft — allowing

the dominant force in the desktop-centric approach to evolve gradually into the major force in network computing. More rapid change would enable the other contenders to provide a viable alternative to a still essentially desktop-oriented Microsoft, thereby establishing market share and making their platform or approach the market standard.

6.1 Interoperability standards

Agreement on an object model and a set of standards for interoperability and common services is essential to the full development of the network computing approach. This is because access to a range of services on the network, in the absence of the ability to install software locally, is essential to the flexibility of the model, i.e. the ability to adequately meet the varying requirements of different users at different times. The standards currently contending in the market place to support the requirement for interoperability are Microsoft's object model DCOM and the OMG's CORBA. Importantly, software is currently available to provide interoperability between the two models and an attempt is currently being made by the OMG to standardise that interoperation.

6.2 Security

Developments in technologies of encryption and authentication will be critical to the realisation of network computing in a variety of ways [16, 17]. Encryption will be necessary to support storage of personal data at remote sites, to authenticate software to be downloaded from or accessed at network servers, to support micro-payment pricing models, and to support authentication of clients accessing 'extranets' of linked organisations.

6.3 Component software

The ability to download a software component offering a specific function (e.g. a spell-checking module) with the ability to interact with other components (perhaps downloaded from other sources) is potentially one of the most powerful aspects of the network computing approach. Components need to be able to exchange data with and react to events generated by other components. Current approaches include OpenDoc, Java Beans and ActiveX.

6.4 Bandwidth

The availability of high-bandwidth network connections is likely to prove less critical a factor in the overall success or failure of the network computing approach than is generally believed. A network computer is clearly dependent on the network to a greater degree than the PC in

that in its pure form (as a diskless client) the NC is not capable of doing anything without a connection to the network. With regard to bandwidth, however, network computing applications are available (e.g. Applix's Java Office Suite) which operate successfully on modem connections of 28.8 kbit/s. The 'thin client' downloaded to the NC handles only rendering of the application on the screen and detection of the position of interface events. All other processing occurs on the server. Therefore, for applications which do not involve video or images, the thin-client approach gives the NC a capability equal to the PC. For applications which involve loading video and images across the network, the requirement for bandwidth is the same (i.e. currently an obstacle) whatever the nature of the client. Availability of high-bandwidth connections would give developers greater flexibility in applications design and would support the delivery of networked multimedia applications.

6.5 Deployment and management of distributed platforms

The issues involved in management of large-scale distributed systems and three-tier client/server architecture are currently not well understood at a practical level. Support tools for management of large distributed systems are not available. This situation is likely to be addressed by suppliers of ORBs and network computing server architectures as commercial deployments become more common.

7. The 'Borneo' network computing application

Experience of the technical issues and the potential benefits of enterprise-scale network computing has been acquired at BT through development of a number of prototypes, in particular a trial network computing platform for the large BT customer service application known as 'ServiceView'. The current system, based on an Oracle Forms application running on PCs and interfacing with the company's MCSS customer service database [18], enables BT customers to directly place and monitor the progress of orders for equipment and services.

To evaluate the potential of network computers in applications of this type BT has implemented a prototype version of the ServiceView application on a network computing platform. The trial system (known as 'Borneo') is a three-tier client/server architecture with Sun JavaStations as client hardware (a client application implemented in Java), an intermediate tier of business objects implemented using the Sun 'NEO' ORB, and a development version of BT's customer service 'MCSS' database [18]. The primary platform components are shown in Fig 3. The Borneo system was implemented using beta versions of the JavaStation, it was trialled at a BT customer

site and demonstrated at the official launch of the JavaStation in New York in November 1996.

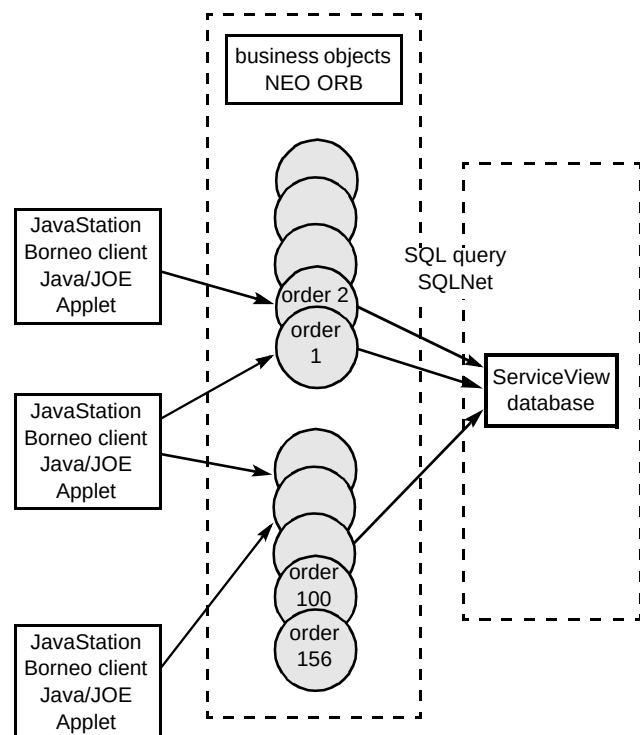


Fig 3 The Borneo platform components.

In the architecture shown in Fig 3, the Borneo client application, implemented as a Java applet, is downloaded on demand from a corporate server to the JavaStation at the customer site. The ability to transparently download the latest versions of the client application, and the 'zero-administration' characteristics of the hardware are of particular benefit given the large number of customer sites at which clients will be installed. In addition, the use of Java enables the client to provide an interface that is more attractive, ergonomic and easier to develop than current interfaces based on technologies such as WWW and Oracle Forms.

The NEO ORB and the Sun Java ORB software (known as JOE) enable the distributed and remote business objects in the intermediate layer of the architecture to be treated by the client application as if they were standard local Java objects. Objects representing components of the business model (such as 'orders' and 'order notifications') can be transparently created, accessed and modified on the remote servers with all the underlying access and communications handled by the JOE/NEO platform. The distributed objects of the intermediate layer use the database as their persistent data store, thereby insulating the client application from the specifics of the large corporate systems with which it may have to interact. This significantly decreases the cost and complexity of application development, particularly where

the client is required to access a variety of different (legacy) systems and where system migration is likely to occur.

8. Conclusions

The market for network computing is very new, and the technology is very dynamic. The realisation of the revenue potential, particularly in the small and medium enterprise and consumer markets, depends on people accepting a more 'on-line' way of life, not just among technology enthusiasts but among 'ordinary' people, including, for example, those who cannot currently program their VCRs. The key to this is in ease of use, both in installing and operating the end-user devices, and in accessing and running applications, and the network computing approach offers real advantages in this area. However, the real potential of network computing will be realised through the use of new forms of 'information appliance', which will in general be of low complexity (but not necessarily low performance) and low cost, and simple to install and use. The latter will often be achieved by tailoring information appliances for use in a specific environment for a specific range of tasks.

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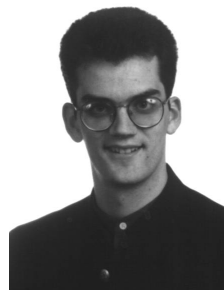


After completing his PhD research in electrical engineering at Essex University and spending a period as a lecturer at Durham University, Mike Revett joined BT in 1973. He initially worked on the development of computer-aided design (CAD) software for VLSI circuits, and during the 1980s he led a group that developed VLSI CAD systems for use within BT, and he managed the BT contribution to a five-year EC-funded project in this area. Since 1990 he has led a group researching and developing information management systems, which has increasingly focused on the use of Internet technology, particularly World Wide Web, to develop intranets — information systems and services to be used within and between companies. He is a Chartered Engineer, and a member of the IEE.



Mike Knul graduated from York University with a BA in English Literature and Philosophy. In 1986 he was awarded an MSc in knowledge-based systems by the University of Essex.

He worked for two years at STC developing knowledge-base applications before moving to BT to work in natural language processing, distributed AI and decision support systems.



Lee Stephens graduated in Computer Science with Electronics at University College Swansea in 1995. He joined BT Laboratories (BTL) in June 1995, and since then has worked on a range of projects involved with the Internet, World Wide Web and network computing. He has been a pioneer user of the Java language at BTL and was a key contributor in the development of the Borneo system, which he has subsequently demonstrated at events in New York, Brussels, Seoul and Sydney.